

11-Day Intensive Defense & Validation Study Plan

Target Oral Defense & System Validation Date: October 14, 2026

Target Defense Date: Wednesday, October 14, 2026	Preparation Period: Oct 4 – Oct 14 (11 Days)
System Name: CDM Integrated Library Management System	Target Focus: Code, Architecture, Manuscript & Demo
Primary Deliverable: 100% Defense Readiness & ISO 25010 Validation	Institutes: ICS, ITE, IBE Programs

Master 11-Day Preparation Schedule

Date & Day	Phase & Focus Area	Daily Tasks & Learning Deliverables	Target Output / Verification
Day 1: Oct 4 (Sunday)	Phase 1: System Tech Stack Mastery	Study overall system architecture: React 19, TypeScript, Tailwind v4, Express.js REST API, SQLite3. Understand how client apps talk to backend API on port 5002.	Able to explain tech stack choices confidently.
Day 2: Oct 5 (Monday)	Phase 1: Database Schema & Relational Design	Master server/db.js: Study 6 tables (books, students, librarians, transactions, reservations, otp_codes), foreign keys, and indexes.	Able to draw ERD and explain foreign keys from memory.
Day 3: Oct 6 (Tuesday)	Phase 1: Security Architecture & Auth	Study JWT token authentication middleware, 10-round salted bcrypt password hashing, and CORS security rules.	Able to explain token flow and password security.
Day 4: Oct 7 (Wednesday)	Phase 2: Backend API Routes Deep-Dive	Walk through server/server.js routes: /api/books, /api/borrow, /api/return, /api/auth/forgot-password, and Email OTP engine.	Understand request payloads and response HTTP status codes.
Day 5: Oct 8 (Thursday)	Phase 2: Desktop SPA & MARC 21 Catalog	Study src/app/App.tsx: MARC 21 tagging standards, useMemo search algorithm, ⌘K keyboard shortcut, BookCard image fallbacks, and Table view.	Able to explain MARC tags and useMemo optimization.
Day 6: Oct 9 (Friday)	Phase 2: Mobile PWA & QR Receipts	Study Mobile PWA portal (mobile-app/), Service Worker offline caching, mobile borrow reservations, and 2D QR Code receipt canvas generation (qrcode).	Demonstrate mobile PWA offline and QR receipt generation.
Day 7: Oct 10 (Saturday)	Phase 3: Manuscript Chapters 1-3 Alignment	Review Chapters 1-3 manuscript: SOP questions 1-5, General/Specific Objectives, Scope & Limitations, and IPO Conceptual Framework.	100% alignment between text manuscript and code functionality.
Day 8: Oct 11 (Sunday)	Phase 3: ISO/IEC 25010 Validation Metrics	Study 5 ISO 25010 evaluation characteristics: Functional Suitability, Usability, Security, Performance Efficiency (< 15ms API), and Reliability.	Ready to present system evaluation metrics to panel.
Day 9: Oct 12 (Monday)	Phase 4: Live System Demo Rehearsal	Conduct full live demonstration run-through: Log in as Librarian -> Search catalog -> Process borrow -> Generate QR receipt -> Test Student Mobile PWA -> Reset password via OTP.	Seamless live system presentation under 10 minutes.
Day 10: Oct 13 (Tuesday)	Phase 4: Mock Panel Q&A; Simulation	Conduct mock defense session with team: Practice answering tricky panel questions on security, SQLite choice, performance, and catalog standards.	Every team member confident with assigned presentation parts.

Day 11: Oct 14 (Wednesday)	ORAL DEFENSE & VALIDATION DAY	Arrive early, boot local servers (Port 5002 backend, Port 5173 desktop, Port 5174 mobile), verify database connection, and deliver a victorious capstone defense!	■ SUCCESSFUL CAPSTONE DEFENSE & VALIDATION!
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Team Role Assignments for Defense Presentation

Defense Role	Primary Focus & Presentation Scope	Files & Artifacts to Master
Team Lead / Presenter	Opening statement, Chapter 1 background, objectives, system live demonstration overview, and concluding remarks.	Manuscript Ch 1, System Live Demo Workflow
System Architect / Backend Lead	Backend REST API routes, JWT security, `bcrypt` password hashing, SQLite database schema, and Email OTP recovery engine.	server/server.js server/db.js
Frontend & Mobile Lead	Desktop SPA interface, MARC 21 cataloging, `useMemo` search/filter algorithm, Mobile PWA portal, and QR receipt verification.	src/app/App.tsx cdm_library_mobile.jsx
Quality Assurance & Research Lead	Chapter 2 literature synthesis, Chapter 3 SDLC methodology, ISO/IEC 25010 software quality evaluation metrics, and statistical results.	Manuscript Ch 2 & 3, ISO 25010 Survey Results